

Vector spaces. Subspaces

K field.

A vector space over K / A K -vector space is an abelian group $(V, +)$ together with a so-called external operation or scalar multiplication.

$$\therefore K \times V \rightarrow V, (k, v) \mapsto k \cdot v$$

left vector space

$$\hookrightarrow \varphi(k, v) = k \cdot v$$

Satisfying the following axioms:

$$\left\{ \begin{array}{l} L_1: k \cdot (v_1 + v_2) = k \cdot v_1 + k \cdot v_2 \quad \begin{array}{l} \nearrow \text{from } V \\ \text{from } K \end{array} \\ L_2: (k_1 + k_2) \cdot v = k_1 \cdot v + k_2 \cdot v \quad \begin{array}{l} \text{from } K \\ \text{from } V \end{array} \\ L_3: (k_1 \cdot k_2) \cdot v = k_1 \cdot (k_2 \cdot v) \quad \begin{array}{l} \text{from } K \\ \text{external op.} \end{array} \\ L_4: 1 \cdot v = v \quad \begin{array}{l} \downarrow \\ \in K \end{array} \end{array} \right.$$

$k_1, k_2, k \in K \rightarrow$ scalars

$v_1, v_2, v \in V \rightarrow$ vectors

$A \times A \rightarrow A$ was an internal operation
 $K \times V \rightarrow V$ is an external operation

Vector space $\sim$ Linear space

Notation: ${}_K V$ or $(V, K, +, \cdot) \rightsquigarrow$ left vector space

Every vector has:

- direction
- orientation
- length
- origin

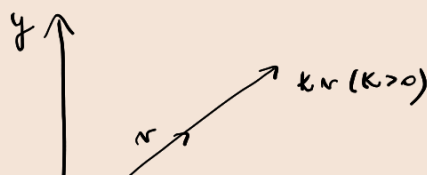
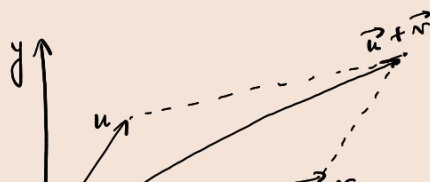
! In a cartesian product, the order of elements does matter!

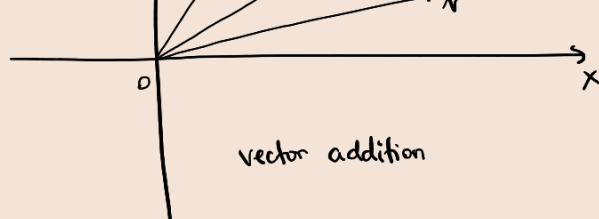
$L_1, L_2 \rightsquigarrow$ distributive laws } sort of, but they are not, since the terms
 $L_3 \rightsquigarrow$ associative law } aren't from the same set.

$$\therefore V \times K \rightarrow V, (v, k) \mapsto v \cdot k$$

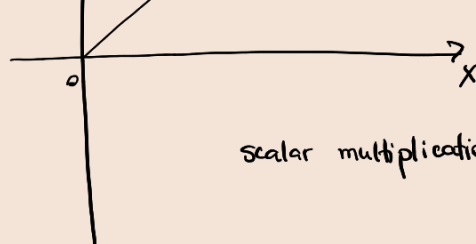
right vector space

${}_K V_2$





vector addition



scalar multiplication

$V_2 \rightarrow$ the abelian group $\begin{cases} \vec{0} \text{ id. elem.} \\ + \rightarrow \text{vectors do have opposites} \\ \text{assoc} \rightarrow \text{a third vector} \end{cases}$
 $\mathbb{R} \rightarrow$ the field

• origin + ending point $\Rightarrow$ you have your vector.

The addition of vectors: $(x, y) + (x', y') = (x + x', y + y')$

The scalar multiplication of vectors: $k(x, y) = (k \cdot x, k \cdot y)$

$(\forall) k \in \mathbb{R}, (\forall) (x, y), (x', y') \in \mathbb{R} \times \mathbb{R}$

$\Rightarrow (\mathbb{R}^2, \mathbb{R}, +, \cdot)$ vector space

$$\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$$

$\Rightarrow$ Generalisation:

$$(x_1, \dots, x_m) + (y_1, \dots, y_m) = (x_1 + y_1, \dots, x_m + y_m)$$

$$k \cdot (x_1, \dots, x_m) = (k \cdot x_1, \dots, k \cdot x_m)$$

$(\forall) (x_1, \dots, x_m), (y_1, \dots, y_m) \in K^m, (\forall) k \in K$

$\Rightarrow (K^m, K, +, \cdot)$ CANONICAL vector space

$$K = \mathbb{Z}_2 \rightarrow \{0, 1\}$$

$m=1 \Rightarrow K$ vector space.

$\mathbb{Q}, \mathbb{R}, \mathbb{C}$ vector spaces

$V = \{0\}$ the zero vector space

A subfield of $K \Rightarrow A$ where $\begin{matrix} "+" \rightarrow K \\ "\cdot" \rightarrow K \end{matrix}$

$\mathbb{Q}, \mathbb{R}, \mathbb{C}$

$(K[x], K, +, \cdot)$ vector space $\begin{cases} \text{usual addition} \\ \text{scalar multiplication} \end{cases}$
 $(\forall) f = a_0 + a_1x + \dots + a_m x^m \in K[x]$
 $k \cdot f = k \cdot a_0 + (k \cdot a_1)x + \dots + (k \cdot a_m)x^m$

usual addition of matrices

$(M_{m,n}(K), K, +, \cdot)$ $\hookrightarrow$ usual scalar multiplication of matrices

$$K^A = \{f \mid f: A \rightarrow K\}$$

$(K^A, K, +, \cdot)$ vector space $\begin{cases} \text{addition: } (f+g)(x) = f(x) + g(x) \\ \text{multip: } (kf)(x) = k \cdot f(x) \end{cases}$

$\hookrightarrow$ Particular: $(\mathbb{R}^{\mathbb{R}}, \mathbb{R}, +, \cdot)$

ii)

i)

$$(1+2) \cdot (3,4) = 3 \cdot (3,4) = (9,4) \quad \times$$

$$1 \cdot (3,4) + 2 \cdot (3,4) = (3,4) + (6,4) = (9,8)$$

Theorem

$\begin{matrix} V \\ n \end{matrix}$

$$a) \quad K \cdot 0 = 0 \cdot v = 0$$

$\swarrow$ vector $\quad \searrow$ scalar

$$\begin{aligned} \hookrightarrow \text{proof: } K \cdot 0 + K \cdot v &= K(0+v) = K \cdot v \quad \leftarrow \text{vectors} \quad \rightarrow K \cdot v \text{ has an inv} \\ &= K \cdot v \quad \leftarrow \\ \Rightarrow K \cdot 0 &= 0 \text{ true} \end{aligned}$$

$$b) \quad K \cdot (-v) = (-K) \cdot v = -K \cdot v$$

$\hookrightarrow$ proof:

$$c) \quad K \cdot (v - v') = K \cdot v - K \cdot v'$$

$\hookrightarrow$ proof:

$$d) \quad (K - K') \cdot v = K \cdot v - K' \cdot v$$

$\hookrightarrow$ proof:

Theorem

$$k^V, k \in K, v \in V$$

$$\text{Then } k \cdot v = 0 \Leftrightarrow k = 0 \text{ or } v = 0.$$

- we have no zero divisors -

$$(\forall) x, y \in K \text{ s.t. } x \cdot y = 0 \Rightarrow x = 0 \text{ or } y = 0.$$

Subspace

$${}_K V, S \subseteq V$$

$$S \text{ is a subspace of } V \text{ if: } \begin{cases} S \neq \emptyset \\ (\forall) v_1, v_2 \in S, v_1 + v_2 \in S \\ (\forall) k \in K, (\forall) v \in S, k \cdot v \in S. \end{cases}$$

$$(\forall) S \leq {}_K V \text{ is a subgroup of } (V, +) \Rightarrow 0 \in S.$$

! Always check if $0 \in S$. If it does not, then S is not a subspace.
0 must always be in a subspace.

Characterisation of subspaces:

$$S \subseteq V, {}_K V$$

$$S \leq V \Leftrightarrow \begin{cases} S \neq \emptyset \text{ (} 0 \in S \text{)} \\ (\forall) k_1, k_2 \in K, (\forall) v_1, v_2 \in S, k_1 \cdot v_1 + k_2 \cdot v_2 \in S. \end{cases}$$

$$\text{If } k_1 = k_2 = 1 \Rightarrow v_1 + v_2 \in S \Rightarrow \text{definition.}$$

Trivial subspaces:

$${}_K V \begin{cases} \rightarrow \{0\} \\ \rightarrow V \end{cases}$$

$$\text{grad } 0 = -\infty.$$

$$\text{Ex I}$$

$$\text{Ex II}$$

$$\text{Ex III}$$

$$-1 = 1 \text{ in } \mathbb{Z}_2$$

$$x \text{ or } \rightarrow \dots$$

$$2 \times x \text{ or } \rightarrow \text{the same}$$

